

Cyber Security Essentials

Beginner's Guide to Ethical Hacking & SOC Fundamentals

1. Introduction

Cyber security protects systems, networks, and data from unauthorized access, attacks, and damage.

2. CIA Triad

Confidentiality, **Integrity**, and **Availability** are the three core security principles.

3. Networking Basics

Learn IP addressing, TCP/IP, DNS, HTTP/HTTPS, firewalls, switches, routers, and ports.

4. Linux Essentials

Understand file permissions, shell commands, package management, and basic Bash scripting.

5. Common Cyber Attacks

Phishing, malware, ransomware, SQL injection, cross-site scripting (XSS), brute-force attacks, and DDoS.

6. OWASP Top 10

Review the most critical web application security risks and secure coding practices.

7. Security Tools

Wireshark, Nmap, Burp Suite, Metasploit, Nessus, and Splunk are widely used for analysis and testing.

8. SOC Analyst Basics

Learn log monitoring, SIEM concepts, alert triage, incident response, and threat intelligence.

9. Incident Response

Preparation, identification, containment, eradication, recovery, and lessons learned.

10. Best Practices

Use MFA, strong passwords, regular updates, backups, least privilege, and user awareness training.

11. Interview Preparation

Study networking, Linux, OWASP, SIEM, incident response, and hands-on labs.

12. Learning Resources

Practice with TryHackMe, Hack The Box, OWASP Juice Shop, and official documentation. Always operate within authorized environments.

Disclaimer: This eBook is for educational and defensive security purposes only. Perform security testing only on systems you own or are explicitly authorized to assess.